

# OG-Core $\times$ CLEWS — The Coupling Mechanisms

A reference: the intuition, then the technical form, for each channel.  
Numbers are the committed M=8 Philippine golden record.

## 1 The coupling in brief

OG-CORE and CLEWS are complementary. CLEWS/OSEMOSYS is a least-cost energy optimiser — rich in technology, dispatch, and emissions, but blind to behaviour: its demand is a fixed input, and it has no households, government budget, or distribution. OG-CORE is an overlapping-generations general-equilibrium economy — rich in households (by age  $s$  and lifetime income  $j$ ), firms ( $M$  industries), and a government that closes its budget — but blind to energy: energy is not a priced input to firm production. They are joined by a *staged soft link* that exchanges a few signals; a single hard model is ruled out because one steps through time period by period while the other plans all years at once.

Three objects organise every mechanism:

- **Signal** — a quantity read from a *solved* CLEWS scenario: a fuel’s marginal cost, the capital a build-out needs, emissions, an implied carbon price.
- **Wedge** — the OG-CORE parameter that carries it: the consumption tax  $\tau^c$ , an industry productivity  $Z_m$ , public investment  $\alpha_I$ , mortality  $\rho(s, t)$ , effective labour  $e(t, s, j)$ .
- **Channel** — one signal mapped to one wedge, with a guard that fires when the mapping is large or ill-posed.

The load-bearing signal is the **energy price**: CLEWS’s marginal cost of supplying one more unit. Fed to OG-CORE, it does two things CLEWS cannot — it lets demand respond to price, and it reveals who bears the cost. Because OG-CORE firms have no energy input, the energy-price channel represents an electricity-price change through electricity’s *two* uses, combined in one transmission: a cost-push on the industries that buy electricity as an input (the contractionary, wage-bearing side), and a recycled consumption wedge on households (the regressive cost-of-living side). The input side is a reduced-form proxy; the fully-structural version — energy as a priced production input — is not yet built.

Throughout, a hat denotes a reform/baseline ratio ( $\hat{x} \equiv x^r/x^b$ ), and  $i_e / m_e$  are the energy *good* / *industry* (Philippines:  $m_e = 2$ ).

## 2 The mechanisms

The channels split by direction: four carry CLEWS signals into OG-CORE, two are policy levers applied once to both models, and two send OG-CORE results back to CLEWS.

## 2.1 Energy price

[CLEWS → OG]

**Intuition.** An electricity-price change hits the economy in two places, because electricity is used in two ways. Industries buy most of it as an *input* (in the Philippine accounts about 73%), so a higher price raises their costs — and, through them, wages and output. Households buy the rest *directly* (about 25%), so it raises the cost of living, hardest for the poor (energy is a larger share of a poor budget). The channel is one complete transmission that carries both.

**Technical.** The reform/baseline electricity-price ratio  $g$  is applied through both uses. (i) *Firm input* — a *cost-push*: each industry  $j$ ’s productivity is cut by electricity’s share  $\phi_j$  of its input costs (from the SAM),  $Z_j \mapsto Z_j/(1 + \phi_j(g - 1))$ , so its output price rises by  $\approx \phi_j(g - 1)$  — the contractionary, wage-bearing side. (ii) *Household consumption* — a *wedge*: a recycled consumption-tax wedge on the energy good, scaled to electricity’s value-share  $\sigma$  of that good,

$$1 + \tau_{i_e}^{c,r} = (1 + \sigma(g - 1))(1 + \tau_{i_e}^{c,b})$$

— the regressive cost-of-living side. Electricity’s own self-use is zeroed in the cost-push so it is not double-counted by the wedge. The input side is a reduced-form proxy (OG-CORE has no energy in production; the structural version is a real inter-industry use matrix, unbuilt). The price  $g$  is CLEWS’s “auto” source — a cost-of-electricity index if available, else the OSEMOSYS commodity-balance marginal cost; for the Philippines it is near-flat, so the whole channel is small.

**Variables.**  $g = \pi^r/\pi^b$ , the reform/baseline electricity-price ratio;  $\phi_j$ , electricity’s share of industry  $j$ ’s input costs;  $\sigma$ , electricity’s value-share of the household energy good  $i_e$ ;  $Z_j$ , industry  $j$ ’s productivity;  $\tau_{i_e}^c$ , the consumption-tax wedge on the energy good ( $r, b$  = reform, baseline).

## 2.2 Carbon price

[policy — both models]

**Intuition.** One carbon price, chosen by policy, applied once on both sides: in the energy model it tilts the build toward cleaner technology; in the economy it raises the price of (household) energy like a tax. It is counted once — never also inferred back from the energy model’s own implied carbon cost.

**Technical.** A price  $cp$  (US\$/tCO<sub>2</sub>) becomes an add-on to the energy-good wedge in OG-CORE, and an emissions-penalty path in CLEWS:

$$\Delta\tau_{i_e}^c = \frac{cp \cdot \text{deflator} \cdot \text{carbon intensity}}{p_{i_e}^b}.$$

Because OG-CORE firms buy no energy, only *household* energy carbon is priced on the macro side; industrial carbon is priced in CLEWS. Revenue is recyclable as transfers ( $\alpha_T$ ). Standalone  $Y_{ss} = -0.040\%$ ; the *level* rides on the uncalibrated unit/deflator bridge, so treat the magnitude as illustrative.

**Variables.**  $cp$ , the carbon price; the deflator maps CLEWS money to the OG-CORE numéraire (currently 1, uncalibrated); carbon intensity, tCO<sub>2</sub> per unit of the energy good;  $\alpha_T$ , the transfer share used to recycle revenue.

## 2.3 Public investment

[CLEWS → OG]

**Intuition.** Building the grid (transmission and distribution) is a public cost. If debt-financed it competes with everyone else for savings, pushing up the interest rate and crowding out private investment; but public capital is also productive, so it lifts output. The net sign depends on which dominates.

**Technical.** The public grid-capex increment (as a share of GDP) is added to the public-investment share  $\alpha_I$  on a finite ramp; it accumulates into public capital, which is productive in the firm’s technology:

$$K_{g,t+1} = (1 - \delta) K_{g,t} + I_{g,t}, \quad I_{g,t} = \alpha_{I,t} Y_t.$$

This is where crowding-out legitimately lives. In the Philippine scenario the extra grid spend is  $\approx 0$  (the transition is generation-side), so the channel is inert here ( $Y_{ss} = +0.000\%$ ); its mechanics scale with any non-zero spend.

**Variables.**  $\alpha_I$ , the public-investment share of output;  $K_g$ , the public capital stock;  $I_g$ , public investment;  $\delta$ , depreciation;  $\gamma_g$ , public capital’s share in production.

## 2.4 Capital intensity

[CLEWS → OG]

**Intuition.** A clean power fleet is capital-heavy (much up-front capex, little fuel). The tempting reading — “so it pulls capital in” — is wrong here. Raising the energy sector’s capital *share* with productivity fixed is like making the sector cheaper to run: competition drives the electricity price down, and because electricity demand barely responds, the sector’s revenue shrinks and it *sheds* capital. So this is a factor-income / energy-price lever, not a capital draw-in.

**Technical.** The reform/baseline ratio of the power fleet’s capital-cost share scales the energy industry’s capital exponent  $\gamma_m$  (Philippines: +12.2%). Under Cobb–Douglas technology ( $\varepsilon = 1$ ) capital’s revenue share *is*  $\gamma_m$ , so firm capital demand is  $K_m = \gamma_m p_m Y_m / \rho_m$ . The lever moves only  $\gamma_m$ , so in the reform/baseline ratio the cost of capital  $\rho_m$  and the return  $r$  cancel, leaving an exact identity:

$$\hat{K}_E = \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E.$$

Measured (energy industry):  $\hat{\gamma}_E = +12.2\%$ ,  $\hat{p}_E = -42.8\%$ ,  $\hat{Y}_E = +20.9\%$ , so  $\hat{K}_E = -22.4\%$  (the identity reproduces it to 0.03%).

**Variables.**  $\gamma_m$ , industry  $m$ ’s capital exponent (its share of value added);  $\rho_m = (r + \delta - \tau_m^b \delta_m^\tau - \tau_m^{\text{inv}} \delta) / (1 - \tau_m^b)$ , the cost of capital;  $p_m, Y_m$ , the industry’s price and output; a hat is a reform/baseline ratio;  $E$  marks the energy industry.

## 2.5 Investment incentive

[policy]

**Intuition.** To actually pull capital *into* energy, lower its cost of capital — for instance with an investment tax credit. That makes energy investment cheaper at the going interest rate, so capital reallocates in. This is a genuinely different lever from capital intensity: capital intensity changes the factor *mix* (the exponent); the credit changes the *cost of capital*. They move energy capital in opposite directions, so they are not two views of the same thing.

**Technical.** An investment tax credit  $\tau_m^{\text{inv}}$  on the energy industry enters the cost of capital directly, where the capital exponent is *absent*:

$$\rho_m = \frac{r + \delta - \tau_m^b \delta \tau_m^\tau - \tau_m^{\text{inv}} \delta}{1 - \tau_m^b}.$$

A higher  $\tau_m^{\text{inv}}$  lowers  $\rho_m$ , so capital demand  $K_m = \gamma_m p_m Y_m / \rho_m$  shifts out at the prevailing  $r$ . It is a defined lever, not part of the Philippine reference run.

**Variables.**  $\tau_m^{\text{inv}}$ , the investment tax credit rate;  $\tau_m^b$ , the business-income tax;  $\delta$ , economic depreciation;  $\delta_m^\tau$ , tax depreciation;  $\rho_m$ , the cost of capital.

## 2.6 Health

[CLEWS  $\rightarrow$  OG]

**Intuition.** Cleaner air saves lives and — more important for output — makes working-age people healthier and more productive. But a 7.7% cut in power-sector PM<sub>2.5</sub> is not 7.7% fewer deaths: energy is one pollution source among many, and deaths fall less than one-for-one as the air clears, so only  $\sim 8\%$  of the cut reaches mortality. The deaths avoided skew elderly (retirees), so the long-run output effect is  $\approx 0$ ; the real macro gain is the working-age productivity channel, which shows up along the transition.

**Technical.** The reform/baseline PM<sub>2.5</sub> emissions change  $\Delta\text{em}$  scales the country’s Global Burden of Disease attributable deaths by a calibrated dose-response  $M$ :

$$\Delta\text{deaths} = D_{\text{PM}} \times M \times \Delta\text{em} = 43,951 \times 0.082 \times (-0.077) \approx -279.$$

*Mortality* shocks the age-specific rate  $\rho(s, t)$  to hit that signed target and recomputes the population; *morbidity* scales the effective-labour profile  $e(t, s, j)$  by a working-age disability shape. Result:  $Y_{ss} = -0.003\%$  but  $+0.19\%$  near-term. The  $\times M$  is what separates  $-279$  from the naïve  $-3,400$ .

**Variables.**  $\Delta\text{em}$ , the reform/baseline PM<sub>2.5</sub> emissions change;  $D_{\text{PM}}$ , GBD ambient-PM<sub>2.5</sub> attributable deaths (43,951, Philippines);  $M \approx 0.082$ , the dose-response (energy mass-share  $\times$  concentration-response elasticity);  $\rho(s, t)$ , mortality by age  $s$  and time  $t$ ;  $e(t, s, j)$ , effective labour by time, age, and income group  $j$ .

## 2.7 Discount rate

[OG  $\rightarrow$  CLEWS]

**Intuition.** The economy has an equilibrium return on capital — its own cost of capital. Handing that to the energy planner means it discounts future costs at a rate the economy actually implies, rather than an arbitrary assumption. A lower return favours capital-heavy, long-lived assets (renewables).

**Technical.** After the reform solves, OG-CORE’s equilibrium market return  $r_p$  (a multi-period mean) is written to CLEWS’s planning discount rate. It is emitted *after* the solve and is one-way: it changes the energy build, whose prices feed back to OG-CORE only once the loop iterates. Inert as a standalone reform.

**Variables.**  $r_p$ , the market portfolio return (OG-CORE’s equilibrium cost of capital); the CLEWS discount rate it sets.

## 2.8 Energy demand

[OG  $\rightarrow$  CLEWS]

**Intuition.** A larger or richer economy needs more energy. Once the economy solves, its activity scales the demand the energy system must meet. In a single pass this barely changes, so the channel is latent — it only bites once the two models iterate.

**Technical.** The reform/baseline activity ratio (industry output  $Y_m$  or consumption  $C_i$ , raised to an elasticity) scales CLEWS's specified annual demand:  $\text{demand}^r = (\hat{Y}_m)^{\varepsilon_D} \text{demand}^b$ . Emitted after the solve; one-way. In one pass  $\hat{Y}_m \approx 1$ , so it is inert by construction.

**Variables.**  $Y_m$ , industry output;  $C_i$ , consumption of good  $i$ ;  $\varepsilon_D$ , the demand-scaling elasticity; the CLEWS specified annual demand it scales.